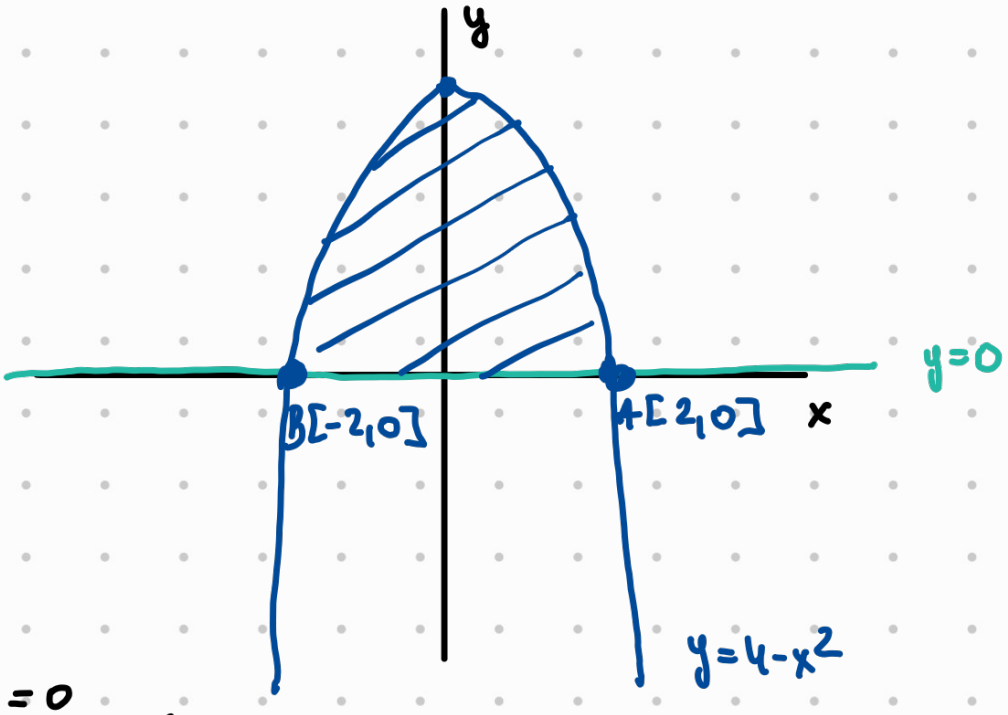
Príklad Nájdite extrémy funkcie $f(x,y) = 4 - 4x - 2y$ na množine $H = \{[x,y] \in R^2 \mid y = 4 - x^2, \; y = 0\}$

1. KROK



Lokálne extrémy:

$f'_x = -4 = 0$

$f'_y = -2 = 0$ ×

2. KROK

$y = 0$

$f(x,y) = 4 - 4x - 2 \cdot 0 = 4 - 4x$

$f' = -4 \neq 0$

$y = 4 - x^2$

$f(x,y) = 4 - 4x - 2(4 - x^2) = 4 - 4x - 8 + 2x^2$

$= 2x^2 - 4x - 4$

$f' = 4x - 4 = 0$

$4x = 4$

$x = 1 \qquad y = 4 - 1^2 = 3$

3. KROK:

$A[2,0] = -4$

$B[-2,0] = 12$ MAX

$C[1,3] = -6$ MIN

$C[1,3]$

Funkcia $f(x,y)$ má v bode $[-2,0]$ maximum s hodnotou 12 a v bode $[1,3]$ minimum s hodnotou -6.

Príklad Nájdite viazané lokálne extrémy funkcie $f(x,y) = 3x^2 + 5y^2 - 2xy$ ak sú viazané podmienkou $x + y = 5$.

$f(x,y) = 3x^2 + 5y^2 - 2xy \qquad g(x,y) = x + y - 5$

1. KROK Lagrangeova rovnica

$L(x,y) = f(x,y) + \lambda g(x,y)$

$L(x,y) = 3x^2 + 5y^2 - 2xy + \lambda(x + y - 5)$

2. KROK

$$\left.\begin{aligned} L'_x &= 6x - 2y + \lambda = 0 \quad (1)\\ L'_y &= 10y - 2x + \lambda = 0 \quad (2)\\ x + y - 5 &= 0 \quad (3) \end{aligned}\right\} \text{ sústava 3 rovníc}$$

(1) $6x - 2y + \lambda = 0$

$2y - 6x = \lambda$

(2) $10y - 2x + 2y - 6x = 0$

$12y - 8x = 0$

$12y = 8x$

$x = \frac{3}{2}y$

(3) $\frac{3}{2}y + y = 5 \quad / \cdot 2$

$3y + 2y = 10$

$5y = 10$

$y = 2$

$A[3,2] \qquad \lambda = -14$

3. KROK:

$$\begin{vmatrix} L''_{xx} & L''_{xy} \\ L''_{yx} & L''_{yy} \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ -2 & 10 \end{vmatrix} = 60 - 4 = 56$$

$56 > 0$ MÁME EXTRÉM

$L''_{xx} = 6 > 0$ MINIMUM

Funkcia má v bode $[3,2]$ minimum s hodnotou ___ .